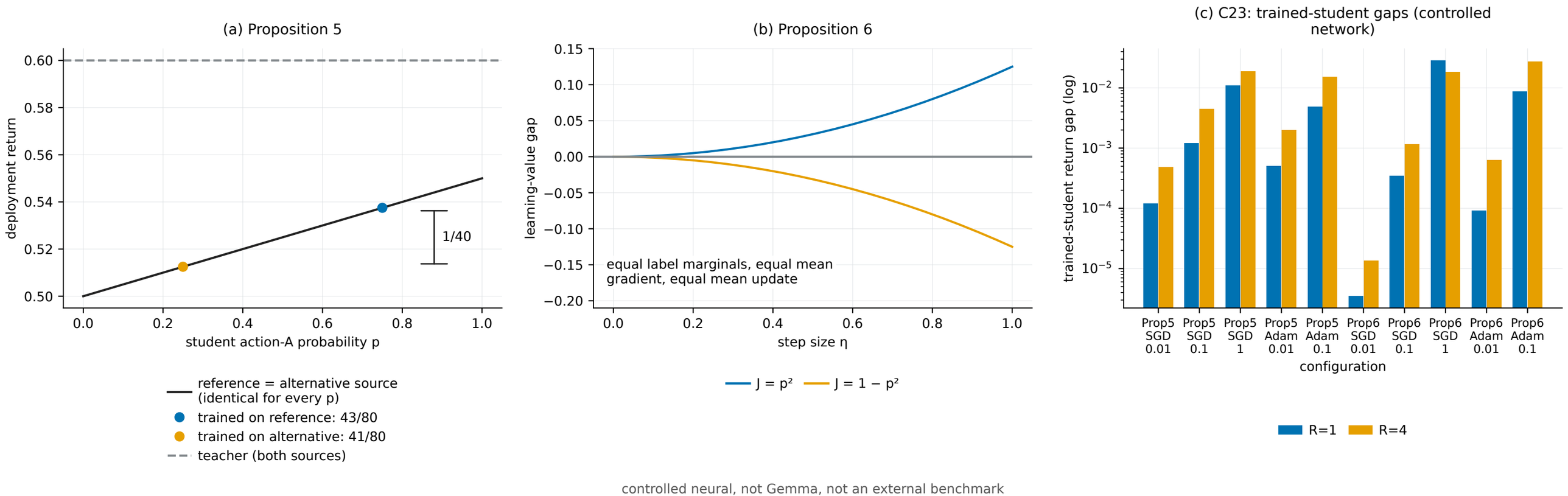
(a) Proposition 5
deployment return
0.60
0.58
0.56
0.54
0.52
0.50
0.0
0.2
0.4
0.6
0.8
1.0
student action-A probability p
1/40
reference = alternative source
(identical for every p)
trained on reference: 43/80
trained on alternative: 41/80
teacher (both sources)
(b) Proposition 6
learning-value gap
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
equal label marginals, equal mean
gradient, equal mean update
step size η
J = p²
J = 1 − p²
(c) C23: trained-student gaps (controlled
network)
trained-student return gap (log)
10⁻²
10⁻³
10⁻⁴
10⁻⁵
Prop5 SGD 0.01
Prop5 SGD 0.1
Prop5 SGD 1
Prop5 Adam 0.01
Prop5 Adam 0.1
Prop6 SGD 0.01
Prop6 SGD 0.1
Prop6 SGD 1
Prop6 Adam 0.01
Prop6 Adam 0.1
configuration
R=1
R=4
controlled neural, not Gemma, not an external benchmark